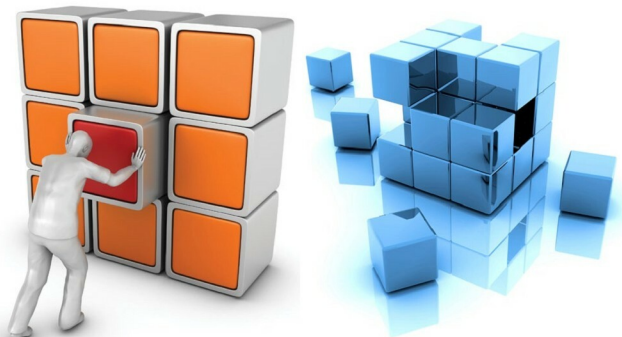


Jumpstart Linux Kernel Module Programming

Here's an introduction to the Linux kernel module, along with an explanation on how to write a simple pluggable module to the Linux kernel and load the module into the existing kernel.



The kernel is the core of any operating system and is responsible for managing system resources. Broadly, the Linux kernel can be of two types.

Monolithic kernels: This is a single executable file in which all the modules are part of the kernel. In order to add anything to the existing kernel, developers have to rebuild the complete kernel and add the new functions.

Modular kernels: Modular kernels provide developers an option to add new functionality to the existing kernel by plugging the new code, also known as 'modules' at run time.

In this article, let's explore how to write a simple pluggable kernel module for the Linux kernel.

What are kernel modules?

Kernel modules are pieces of code, which can be loaded and unloaded from a kernel, on demand. A Linux Kernel Module (LKM) can be added at run time without even requiring a reboot or even a rebuild of the running kernel. The LKM will have a `.ko` extension.

The LKM will act as the interface between a user space application and the Linux kernel. Any request to access the

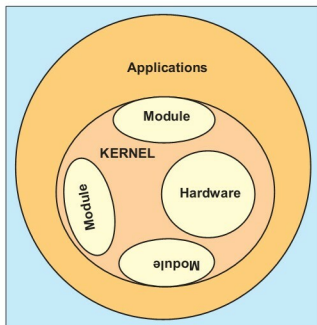


Figure1: The interface between the application and the kernel